Investment Research Report

NASDAQ-100 Optimal Portfolio vs. D-Wave Quantum Inc. (QBTS)

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Report Date: May 25, 2026 | Classification: Internal Research

Executive Summary

This report presents a quantitative comparison between a Monte Carlo-optimized NASDAQ-100 portfolio and D-Wave Quantum Inc. (NYSE: QBTS) over the period January 2 through May 22, 2026. Using a survivorship bias-corrected universe of 104 tickers, 10,000 simulated portfolio allocations were evaluated to identify the maximum Sharpe ratio portfolio. The resulting optimal portfolio is benchmarked against QBTS on key risk-adjusted performance metrics including total return, annualized volatility, Sharpe ratio, maximum drawdown, and Calmar ratio.

The analysis finds that the diversified NASDAQ-100 optimal portfolio substantially outperforms QBTS on every risk-adjusted metric over the study period. While QBTS demonstrates speculative upside potential consistent with early-stage quantum computing equities, its risk profile is incompatible with a core portfolio allocation for most investor mandates.

Methodology

Universe Construction. Current NASDAQ-100 constituents were sourced from Wikipedia and diffed against the historical composition as of January 1, 2026 using the Wikipedia revision history API. This survivorship bias correction identified 3 additions (LITE, SNDK, WMT) and 3 removals (AZN, CSGP, TEAM), producing a full analysis universe of 104 tickers.

Portfolio Optimization. Daily adjusted closing prices were retrieved via yfinance for all 104 tickers. Monte Carlo simulation (10,000 iterations) generated random long-only weight vectors summing to 1.0. Annualized return and volatility were computed for each simulation (252 trading-day convention). The optimal portfolio was selected as the allocation maximizing the Sharpe ratio using a risk-free rate of 4.3% (current U.S. T-bill rate).

QBTS Benchmark. D-Wave Quantum price data was sourced for the identical date window and aligned to the same trading-day index via index intersection to ensure comparability. All return series are computed as log-equivalent simple daily returns.

Performance Summary

Metric	Optimal NASDAQ Portfolio	QBTS (D-Wave Quantum)
Total Return (YTD)	**+26.36%**	+4.51%
Annualized Return	**62.25%**	70.12%
Annualized Volatility	**16.76%**	112.51%
Sharpe Ratio (rf = 4.3%)	**3.46**	0.59
Max Drawdown	**-6.00%**	-58.49%

Metric	Optimal NASDAQ Portfolio	QBTS (D-Wave Quantum)
Calmar Ratio	**10.38**	1.20
Return Correlation	**0.507**	0.507

Portfolio Analysis

Optimal NASDAQ-100 Portfolio.

The Monte Carlo-optimized portfolio delivered a total return of 26.36% over the approximately five-month study period, equating to a 62.25% annualized return. Critically, this performance was achieved with an annualized volatility of only 16.76% — well within the range of typical large-cap equity portfolios. The maximum drawdown of -6.00% reflects strong downside discipline, and the Calmar ratio of 10.38 indicates the portfolio generated over ten dollars of annualized return for every dollar of peak-to-trough loss. The risk-adjusted Sharpe ratio of 3.46 is exceptional and reflects the benefit of diversification across 104 large-cap NASDAQ constituents combined with return-maximizing weight allocation.

D-Wave Quantum Inc. (QBTS).

QBTS delivered a total return of 4.51% over the same period — a fraction of the optimal portfolio's return. While the annualized return figure of 70.12% may appear competitive in isolation, it masks an extraordinarily volatile return path. Annualized volatility of 112.51% — nearly seven times that of the NASDAQ portfolio — produced a maximum drawdown of -58.49%, meaning investors experienced a loss of more than half their capital from peak before any recovery. The Sharpe ratio of 0.59 confirms that QBTS does not adequately compensate investors for the risk assumed. The Calmar ratio of 1.20 further underscores this disparity in risk-adjusted efficiency.

Correlation and Diversification.

The return correlation of 0.507 between the two series indicates a moderate positive relationship — QBTS moves broadly in the direction of the NASDAQ, consistent with its classification as a technology-adjacent equity. However, the significantly higher idiosyncratic volatility in QBTS means this correlation provides limited diversification benefit relative to holding the NASDAQ portfolio alone. Adding QBTS as a satellite position would increase total portfolio volatility without a proportionate improvement in expected return.

> ***"The NASDAQ-100 optimal portfolio delivers a Sharpe ratio nearly 6x that of QBTS (3.46 vs. 0.59) while limiting drawdown to -6% versus -58% for QBTS — a decisive advantage in risk-adjusted efficiency over the study period."***

Investment Recommendation

NASDAQ-100 OPTIMAL PORTFOLIO RATING: OVERWEIGHT	QBTS (D-WAVE QUANTUM) RATING: SPECULATIVE / HOLD

Core Allocation — NASDAQ-100 Optimal Portfolio (Overweight). The diversified NASDAQ-100 portfolio, optimized for maximum Sharpe ratio using Monte Carlo simulation, is recommended as the primary equity allocation. Its combination of strong absolute returns (26.36% YTD), low volatility (16.76% annualized), and minimal drawdown (-6.00%) represents a highly efficient risk-return tradeoff. This portfolio benefits from broad sector exposure across 104 large-cap constituents, systematic weight optimization, and survivorship bias-corrected construction — characteristics that underpin durable performance across varying market regimes.

Satellite Allocation — QBTS (Speculative / Hold). D-Wave Quantum is not recommended as a core portfolio holding given its extreme volatility profile and poor risk-adjusted return metrics. However, for investors with a defined speculative allocation budget and a multi-year investment horizon aligned with the commercialization of quantum computing, a small satellite position of 1–3% of total portfolio value may be appropriate. At this sizing, QBTS's volatility contribution to the total portfolio remains bounded while preserving exposure to potential asymmetric upside if D-Wave achieves key technical or commercial milestones.

Alternative Quantum Exposure. Investors seeking quantum computing thematic exposure with a more manageable risk profile may consider diversified quantum-focused ETFs (e.g., Defiance Quantum ETF — QTUM), which spread single-stock risk across multiple early-stage and established quantum-adjacent names. This approach captures sector beta with meaningfully lower idiosyncratic drawdown risk than a single-name QBTS position.

Key Risk Factors

Concentration Optimization Risk	&Monte Carlo weight optimization is backward-looking. Weights derived from a 5-month window may not generalize to future return regimes. Out-of-sample performance should be monitored and weights rebalanced periodically.
QBTS Fundamental Risk	D-Wave Quantum remains pre-profit with commercially unproven quantum advantage at scale. The stock is highly sensitive to funding cycles, technology milestones, and competitive developments from better-capitalized peers (IBM, Google, IonQ).
Macro & Rate Risk	The 4.3% risk-free rate assumption reflects current T-bill levels. A significant change in the rate environment would alter the Sharpe calculations and the relative attractiveness of both instruments versus fixed income alternatives.
Short Study Period	The 98-trading-day analysis window is insufficient to draw statistically robust conclusions about long-run performance. Results should be validated against longer historical periods before informing capital allocation decisions.

Conclusion

The quantitative evidence is unambiguous within the study period: the Monte Carlo-optimized NASDAQ-100 portfolio dominates QBTS on every material risk-adjusted performance dimension. A Sharpe ratio of 3.46 versus 0.59, a maximum drawdown of -6% versus -58%, and a Calmar ratio of 10.38 versus 1.20 collectively indicate that the diversified portfolio provides superior compensation for risk. QBTS remains a high-conviction speculative instrument for investors with the risk tolerance and investment thesis to support it, but should not displace a well-constructed, optimized equity portfolio as a primary allocation.

This analysis demonstrates the practical value of applying data science methodologies — survivorship bias correction, Monte Carlo simulation, and rigorous risk-adjusted metrics — to portfolio construction decisions. Continued iteration with longer time horizons, factor-based weighting, and out-of-sample validation will further strengthen the analytical foundation of this work.
